

ASSIGNMENT- 8.1

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HALL TICKET NO - 2303A51017

BATCH-01

TASK-01:

The screenshot displays a Visual Studio Code editor interface. The Explorer panel on the left shows a project named 'AI-ASSISTED CODE' containing various task files. The main editor window displays the content of 'task8-1.py', which is a Python script designed to validate passwords based on specific criteria. The script includes comments, imports, a function definition, and test cases. The bottom panel shows the 'PROBLEMS' view, which contains a traceback indicating an 'AssertionError' during the execution of the test cases.

```
task8-1.py > ...
1 #generate a python code for at least 3 assert test cases for is_strong_password(password)
2 #Password must have at least 8 characters.
3 # Must include uppercase, lowercase, digit, and special character.
4 # Must not contain spaces.
5 import re
6 def is_strong_password(password):
7     if len(password) < 8: return False
8     if not re.search(r'[A-Z]', password): return False
9     if not re.search(r'[a-z]', password): return False
10    if not re.search(r'[0-9]', password): return False
11    if not re.search(r'[@$!%*?&]', password): return False
12    if ' ' in password: return False
13    return True
14 # Test cases
15 assert is_strong_password("Abcd@123") == True
16 assert is_strong_password("abcd123") == False
17 assert is_strong_password("ABCD@1234") == True
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-1.py
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/doctest.py", line 2811, in <module>
    sys.exit(_test())
             ~~~~^
  File "/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/doctest.py", line 2799, in _test
    m = _import_(filename[-3])
        ~~~~~~^
  File "/Users/ishamadeeha/Documents/AI-assisted code/task8-1.py", line 17, in <module>
    assert is_strong_password("ABCD@1234") == True
           ~~~~~~^
AssertionError
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-2.py
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

Ln 17, Col 47 Spaces: 4 UTF-8 LF Python 3.11.5 Go Live

TASK-02:

The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files task5-9.py through task8-15.py. The main editor displays a Python script named 'task8-2.py' with the following code:

```
1 #generate a python code for at least 3 assert test cases for a classify_number(n) function
2 #Classify numbers as Positive, Negative, or Zero
3 #Handle invalid inputs like strings and None.
4 #Include boundary conditions (-1, 0, 1).
5 def classify_number(n):
6     if n is None:
7         return "Invalid input: None is not a number"
8     if isinstance(n, str):
9         return "Invalid input: String is not a number"
10    if n > 0:
11        return "Positive"
12    elif n < 0:
13        return "Negative"
14    else:
15        return "Zero"
16    # Test cases
17    assert classify_number(10) == "Positive"
18    assert classify_number(-5) == "Negative"
19    assert classify_number(0) == "Zero"
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
Python + Python 3.11.5 Go Live
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/Library/Frameworks/Python.framework/Versions/3.11/Lib/python3.11/doctest.py", line 2811, in <module>
    sys.exit(_test())
  File "/Library/Frameworks/Python.framework/Versions/3.11/Lib/python3.11/doctest.py", line 2799, in _test
    m = _import_(filename[:3])
  File "/Users/ishamadeeha/Documents/AI-assisted code/task8-1.py", line 17, in <module>
    assert is_strong_password("ABCD@1234") == True
AssertionError
```

TASK-03:

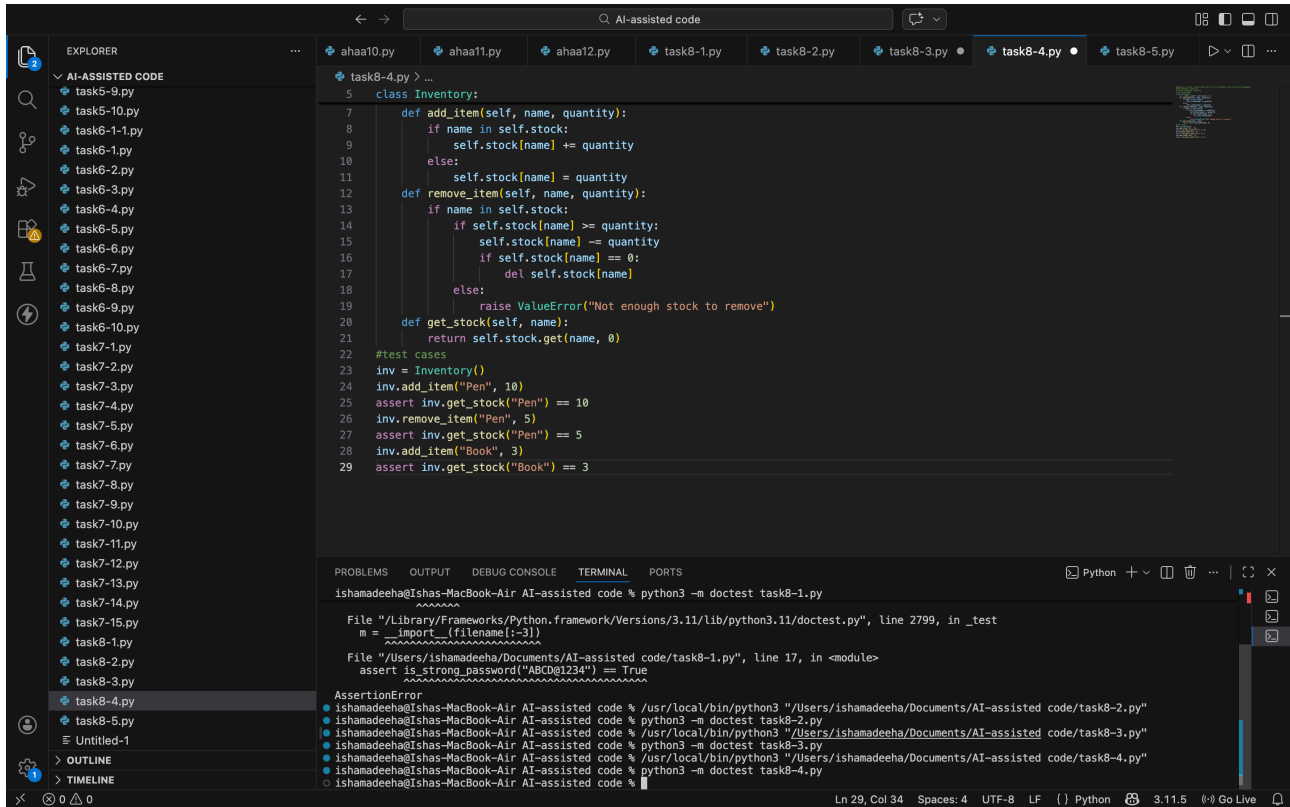
The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files task5-9.py through task8-15.py. The main editor displays a Python script named 'task8-3.py' with the following code:

```
1 #generate at least 3 assert test cases for is_anagram(str1, str2)
2 #Ignore case, spaces, and punctuation.
3 # Handle edge cases (empty strings, identical words).
4 import re
5 def is_anagram(str1, str2):
6     # Remove spaces and punctuation, and convert to lowercase
7     str1_clean = re.sub(r'[\W_]+', '', str1).lower()
8     str2_clean = re.sub(r'[\W_]+', '', str2).lower()
9
10    # Check if the sorted characters of both strings are the same
11    return sorted(str1_clean) == sorted(str2_clean)
12    # Test cases
13    assert is_anagram("listen", "silent") == True
14    assert is_anagram("hello", "world") == False
15    assert is_anagram("Dormitory", "Dirty Room") == True
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
Python + Python 3.11.5 Go Live
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/Library/Frameworks/Python.framework/Versions/3.11/Lib/python3.11/doctest.py", line 2811, in <module>
    sys.exit(_test())
  File "/Library/Frameworks/Python.framework/Versions/3.11/Lib/python3.11/doctest.py", line 2799, in _test
    m = _import_(filename[:3])
  File "/Users/ishamadeeha/Documents/AI-assisted code/task8-1.py", line 17, in <module>
    assert is_strong_password("ABCD@1234") == True
AssertionError
ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-2.py
ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-3.py
```

TASK-04:



```
task8-4.py > ...
5 class Inventory:
7     def add_item(self, name, quantity):
8         if name in self.stock:
9             self.stock[name] += quantity
10        else:
11            self.stock[name] = quantity
12    def remove_item(self, name, quantity):
13        if name in self.stock:
14            if self.stock[name] >= quantity:
15                self.stock[name] -= quantity
16                if self.stock[name] == 0:
17                    del self.stock[name]
18            else:
19                raise ValueError("Not enough stock to remove")
20    def get_stock(self, name):
21        return self.stock.get(name, 0)
22
23 #test cases
24 inv = Inventory()
25 inv.add_item("Pen", 10)
26 assert inv.get_stock("Pen") == 10
27 inv.remove_item("Pen", 5)
28 assert inv.get_stock("Pen") == 5
29 inv.add_item("Book", 3)
30 assert inv.get_stock("Book") == 3
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-1.py

File "/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/doctest.py", line 2799, in _test
m = import_(filename[:3])

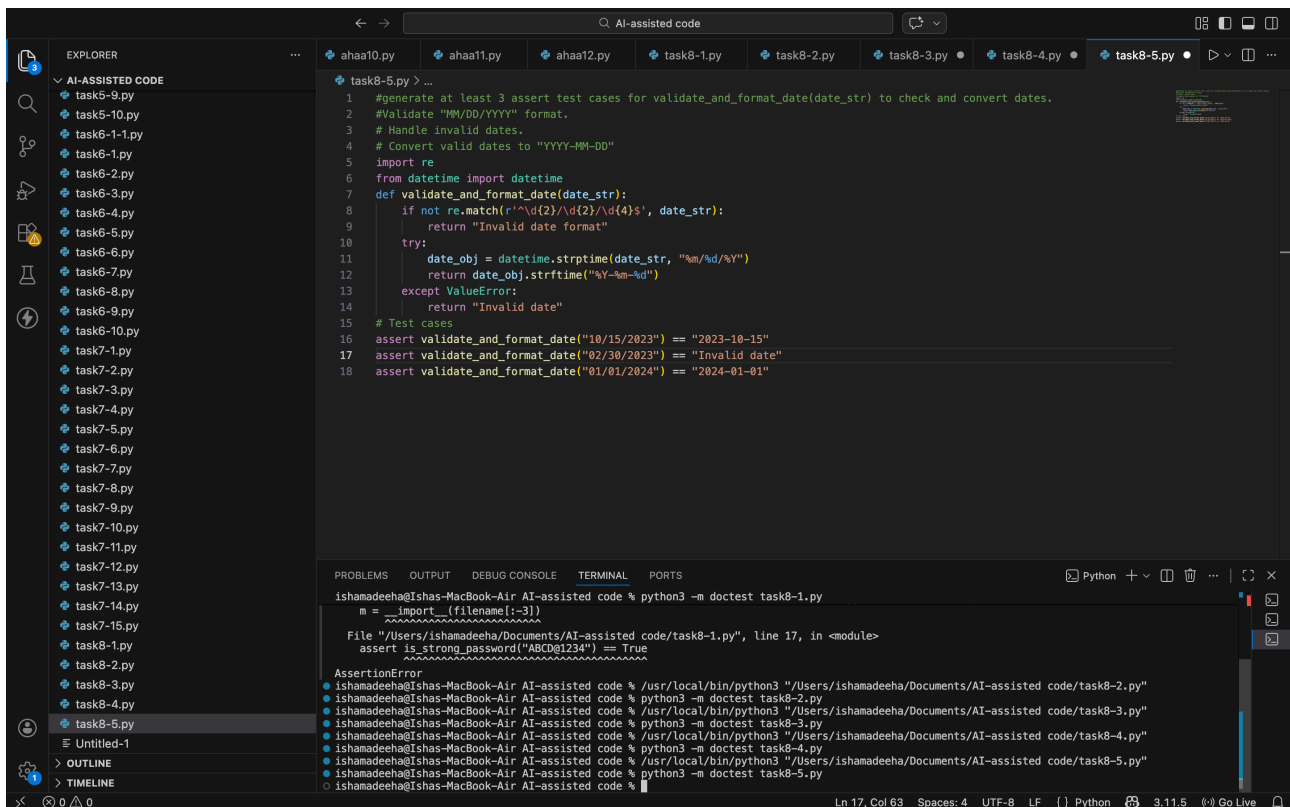
File "/Users/ishamadeeha/Documents/AI-assisted code/task8-1.py", line 17, in <module>
assert is_strong_password("ABCD@1234") == True

AssertionError

- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-2.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-3.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-3.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-4.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-4.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code %

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TASK-05:



```
task8-5.py > ...
1 #generate at least 3 assert test cases for validate_and_format_date(date_str) to check and convert dates.
2 #Validate "MM/DD/YYYY" format.
3 # Handle invalid dates.
4 # Convert valid dates to "YYYY-MM-DD"
5 import re
6 from datetime import datetime
7 def validate_and_format_date(date_str):
8     if not re.match(r'^\d{2}/\d{2}/\d{4}$', date_str):
9         return "Invalid date format"
10    try:
11        date_obj = datetime.strptime(date_str, "%m/%d/%Y")
12        return date_obj.strftime("%Y-%m-%d")
13    except ValueError:
14        return "Invalid date"
15
16 # Test cases
17 assert validate_and_format_date("10/15/2023") == "2023-10-15"
18 assert validate_and_format_date("02/30/2023") == "Invalid date"
19 assert validate_and_format_date("01/01/2024") == "2024-01-01"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-1.py

m = import_(filename[:3])

File "/Users/ishamadeeha/Documents/AI-assisted code/task8-1.py", line 17, in <module>
assert is_strong_password("ABCD@1234") == True

AssertionError

- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-2.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-2.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-3.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-3.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-4.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-4.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task8-5.py"
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m doctest task8-5.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code %

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